

HBase & Hive Integration Case Study - Complete Solution

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Objective:

Create an HBase table, insert data, then create an external Hive table mapped to HBase, and query the data to retrieve all logs where the service is "miner".

Part 1: HBase Operations

1) Create a namespace named bigdata

```
create_namespace 'bigdata'
```

2) Create a table named AnalysisLog_ID with column family LogInfo

```
create 'bigdata:AnalysisLog_ID', 'LogInfo'
```

3) Insert data into the HBase table

```
put 'bigdata:AnalysisLog_ID', '001', 'LogInfo:service', 'FI01Miner01'
put 'bigdata:AnalysisLog_ID', '001', 'LogInfo:message', 'miner info'

put 'bigdata:AnalysisLog_ID', '002', 'LogInfo:service', 'FI01Miner02'
put 'bigdata:AnalysisLog_ID', '002', 'LogInfo:message', 'zookeeper info'

put 'bigdata:AnalysisLog_ID', '003', 'LogInfo:service', 'FI01Miner01'
put 'bigdata:AnalysisLog_ID', '003', 'LogInfo:message', 'miner error'

put 'bigdata:AnalysisLog_ID', '004', 'LogInfo:service', 'FI01Miner02'
put 'bigdata:AnalysisLog_ID', '004', 'LogInfo:message', 'miner error'

put 'bigdata:AnalysisLog_ID', '005', 'LogInfo:service', 'FI01Miner02'
put 'bigdata:AnalysisLog_ID', '005', 'LogInfo:message', 'spark info'

put 'bigdata:AnalysisLog_ID', '006', 'LogInfo:service', 'FI01Miner01'
put 'bigdata:AnalysisLog_ID', '006', 'LogInfo:message', 'zookeeper debug'

put 'bigdata:AnalysisLog_ID', '007', 'LogInfo:service', 'FI01Miner03'
put 'bigdata:AnalysisLog_ID', '007', 'LogInfo:message', 'zookeeper debug'

put 'bigdata:AnalysisLog_ID', '008', 'LogInfo:service', 'FI01Miner02'
put 'bigdata:AnalysisLog_ID', '008', 'LogInfo:message', 'miner info'

put 'bigdata:AnalysisLog_ID', '009', 'LogInfo:service', 'FI01Miner01'
put 'bigdata:AnalysisLog_ID', '009', 'LogInfo:message', 'miner info'
```

4) Verify data insertion

```
scan 'bigdata:AnalysisLog_ID'
```

HBase Shell

Use "help" to get list of supported commands.

Use "exit" to quit this interactive shell.

For Reference, please visit: <http://hbase.apache.org/2.0/book.html#shell>

Version 2.3.3, r3e4bf4bee3a08b25591b9c22fea0518686a7e834, Wed Oct 28 06:36:25 UTC 2020

Took 0.0011 seconds

hbase(main):001:0> get 'bigdata:AnalysisLog_ID', '001'

COLUMN	CELL
LogInfo:message	timestamp=2026-05-31T21:12:16.317, value=miner info
LogInfo:service	timestamp=2026-05-31T21:12:16.192, value=FI01Miner01

1 row(s)

Took 0.8468 seconds

hbase(main):002:0> scan 'bigdata:AnalysisLog_ID'

ROW	COLUMN+CELL
001	column=LogInfo:message, timestamp=2026-05-31T21:12:16.317, value=miner info
001	column=LogInfo:service, timestamp=2026-05-31T21:12:16.192, value=FI01Miner01
002	column=LogInfo:message, timestamp=2026-05-31T21:12:16.448, value=zookeeper info
002	column=LogInfo:service, timestamp=2026-05-31T21:12:16.395, value=FI01Miner02
003	column=LogInfo:message, timestamp=2026-05-31T21:12:16.516, value=miner error
003	column=LogInfo:service, timestamp=2026-05-31T21:12:16.492, value=FI01Miner01
004	column=LogInfo:message, timestamp=2026-05-31T21:12:16.595, value=miner error
004	column=LogInfo:service, timestamp=2026-05-31T21:12:16.569, value=FI01Miner02
005	column=LogInfo:message, timestamp=2026-05-31T21:12:16.657, value=spark info
005	column=LogInfo:service, timestamp=2026-05-31T21:12:16.635,

hbase(main):003:0> describe 'bigdata:AnalysisLog_ID'

Table bigdata:AnalysisLog_ID is ENABLED

bigdata:AnalysisLog_ID

COLUMN FAMILIES DESCRIPTION

{NAME => 'LogInfo', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}

1 row(s)

Part 2: Hive Operations

1) **Start Hive Shell** → hive

2) **Fix Metastore Issues (if any)** → If you encounter metastore errors:

Exit Hive first, then run:

```
rm -rf /home/bigdata/metastore_db
```

```
rm -f /home/bigdata/derby.log
```

```
cd /home/bigdata/hive/bin
```

```
./schematool -dbType derby -reinitSchema
```

```
hive
```

```
(base) [bigdata@localhost ~]$ rm -rf /home/bigdata/metastore_db
(base) [bigdata@localhost ~]$ rm -f /home/bigdata/derby.log
(base) [bigdata@localhost ~]$ rm -f /tmp/derby.log
(base) [bigdata@localhost ~]$ ps aux | grep -i hive | grep -v grep
(base) [bigdata@localhost ~]$ cd /home/bigdata/hive/bin
(base) [bigdata@localhost bin]$ ./schematool -dbType derby -initSchema
Metastore connection URL:          jdbc:derby::databaseName=metastore_db;create=true
Metastore Connection Driver :      org.apache.derby.jdbc.EmbeddedDriver
Metastore connection User:         APP
Starting metastore schema initialization to 2.1.0
Initialization script hive-schema-2.1.0.derby.sql
Initialization script completed
schemaTool completed
(base) [bigdata@localhost bin]$ hive
```

3) **Create Hive Table Mapped to HBase**

```
CREATE EXTERNAL TABLE Log_hive_ID (
    rowkey string,
    service string,
    message string
)
STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'
WITH SERDEPROPERTIES (
    "hbase.columns.mapping" = ":key, LogInfo:service, LogInfo:message"
)
TBLPROPERTIES (
    "hbase.table.name" = "bigdata:AnalysisLog_ID"
);
```

4) Verify the Table Creation

```
SHOW TABLES;
```

```
DESCRIBE Log_hive_ID;
```

```
hive> SHOW TABLES;
OK
log_hive_id
Time taken: 0.042 seconds, Fetched: 1 row(s)
hive> DESCRIBE Log_hive_ID;
OK
rowkey                string
service                string
message                string
Time taken: 0.028 seconds, Fetched: 3 row(s)
hive> █
```

5) Query Data

```
SELECT * FROM Log_hive_ID WHERE service LIKE '%Miner%';
```

```
hive> CREATE EXTERNAL TABLE Log_hive_ID (
>   rowkey string,
>   service string,
>   message string
> )
> STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'
> WITH SERDEPROPERTIES (
>   "hbase.columns.mapping" = ":key, LogInfo:service, LogInfo:message"
> )
> TBLPROPERTIES (
>   "hbase.table.name" = "bigdata:AnalysisLog_ID"
> );
OK
Time taken: 2.063 seconds
hive> SELECT * FROM Log_hive_ID LIMIT 5;
OK
001    FI01Miner01    miner info
002    FI01Miner02    zookeeper info
003    FI01Miner01    miner error
004    FI01Miner02    miner error
005    FI01Miner02    spark info
Time taken: 1.721 seconds, Fetched: 5 row(s)
hive> SELECT * FROM Log_hive_ID WHERE message LIKE '%miner%';
OK
001    FI01Miner01    miner info
003    FI01Miner01    miner error
004    FI01Miner02    miner error
008    FI01Miner02    miner info
009    FI01Miner01    miner info
Time taken: 0.52 seconds, Fetched: 5 row(s)
hive> █
```